

# MIDUS 3 PROJECT 5 NEUROSCIENCE DATA README

September 2026

## A. Changes to the Dataset and Documents

This September 2026 update contains the revised M3P5 dataset with 231 cases and 3472 variables. The prior version of this dataset, M3\_P5\_DATA\_N231\_20260206, contained 3472 variables and was updated in February 2026.

September 2026 dataset update includes:

- Corrections/updates to variables previously shared
  - Updated value labels for HRV filter variables **[C5V1B]** **[C5V1Q]** **[C5V2B]** **[C5V2Q]** to add a label to the missing value of 9998.
  - Corrected corrugator filter variable **[C5C]** to update subjects who were incorrectly marked as 'missing' and changed to 'useable data'.

February 2026 dataset update includes:

- New variables:
  - Computed measures from a new brain-based rate of aging algorithm: the Dunedin Pace of Aging Calculated from NeuroImaging (DunedinPACNI) **[C5EBD]**.
  - Self-reported neurological condition **[C5IN]**
  - Start time of the structural **[C5ITS]** and functional MRI scans **[C5ITF]**
  - Added data quality filter for startle eyeblink data **[C5BF]**
  - Measures of heart rate variability (HRV) **[C5V...]**
  - Additional CANTAB Intra-Extra Dimensional Shift (IED) variables to help interpret scores for subjects who failed the test before completing all stages. Including variables for stage errors **[C5NISE...]**, stage trials **[C5NIST...]**, and 4 computed variables adjusting for test terminating early due to too many incorrect responses **[C5NIC05 – C5NIC08]**
- Corrections/updates to variables previously shared
  - Updated missing values for CANTAB **[C5N...]** to differentiate missing data (i.e., never collected) as opposed to inapplicable data (e.g., failing the task early, no incorrect responses, etc.)
  - Correction to brainage estimated using the algorithm from Yin et al., 2023 - PMID: 36595679 **[C5EBP]** - adding 6 missing values and removing value for 1 subject's data determined too poor quality to compute
  - Cube and paper task corrections 2 subjects **[C5DCB]** corrected for 1 subject and **[C5DR]** corrected for 1 subject
  - Added missing values for 1 subject startle eyeblink **[C5B...]** and Psychophysiology Emotional Response Task reaction time **[C5R...]** and accuracy **[C5A...]**

February 2026 update also includes a reorganization and renaming of documentation to make the documentation more user friendly. A description of these documents is outlined in section D.

## B. What Data Files Are Available?

The MIDUS 3 (M3) Project 5 (P5) Neuroscience dataset is an SPSS file containing 231 cases from the Main and Milwaukee samples.

- *M3\_P5\_DATA\_N231\_20260910*

This file is comprised of:

- self-report measures of emotion, emotion regulation, anxiety, and empathy
- psychophysiological measures including corrugator and zygomatic facial electromyography and eyeblink startle magnitude of emotional reactivity and recovery in response to the presentation of negative, neutral, and positive pictures
- heart rate variability measures collected both in and out of the MRI scanner
- reaction time and accuracy measures obtained during the psychophysiology Emotional Response Task
- cognitive data obtained via pen-and-paper tasks and from CANTAB (<http://www.cambridgecognition.com/>) cognitive research software
- atlas-based extracted brain measures derived from BRAVO T1-weighted magnetic resonance imaging (MRI) scans, including measures of brain structure (volume, cortical thickness, cortical curvature, and cortical surface area) calculated using FreeSurfer software (v6.0.0), hippocampal subfield and amygdala nuclei segmentations calculated using FreeSurfer's hippocampal module (v7.1.0) and
- brain-based aging algorithms derived from BRAVO T1-weighted magnetic resonance imaging (MRI) scans and calculated using multiple publicly shared algorithms and deep learning models
- atlas-based extracted brain measures of white-matter microstructure derived from diffusion-weighted imaging (DWI) scans

### **C. What is the Structure of the P5 Dataset?**

Variables have been named according to MIDUS variable naming and coding conventions. All variables include labels to aide interpretation. Value labels have been applied where appropriate.

Variable naming conventions are described in: *M3\_P5\_VARIABLE\_NAMES\_20260910*

The third character of the variable name is a letter that identifies the type, or name, of the instrument used to collect the data:

- S = Self-reports
- B = Startle Eyeblink
- C = Corrugator EMG
- L = Zygomaticus EMG
- K = Heart Rate Variability (QRSTool and CMetX – used in MR1 only)
- V = Heart Rate Variability (NeuroKit2 – used in M3)
- R = Psychophysiology Emotional Response Task - Response Times
- A = Psychophysiology Emotional Response Task - Response Accuracy
- N = CANTAB Cognitive measures
- D = Cube & Paper Test
- F = Free Recall
- T = Picture Ratings
- P = Participant Characteristics
- H = Handedness
- O = Hearing Test
- I = General MRI Information
- E = Extracted Structural Brain Measurements
- W = Extracted Diffusion Weighted Imaging Measurements

*Remaining characters differ to build discrete variables, but all within the given measure type set by character 3.*

## D. What Additional Documentation Files Are Available?

### M3P4 Overview (MIDUS 3 Biomarker Project documentation)

Participation in the Neuroscience Project was contingent upon also participating in the Biomarker Project - see the MIDUS 3 Biomarker Project documentation for basic information about the sample and recruitment

### M3\_P5 DOCUMENTATION OF PROCEDURES 20260910

An overview of our data collection protocol and the timing of the measures collected during the psychophysiology visit and neuroimaging visit. Also including changes in data collection due to COVID-19

*Note: this document is NEW for the M3 P5 February 2026 data sharing update.*

### M3\_P5 DOCUMENTATION OF BEHAVIORAL COGNITIVE 20260910

Information regarding behavioral and cognitive assessments collected during the psychophysiology visit and neuroimaging visit

*Note: this document is NEW for the M3 P5 February 2026 data sharing update.*

### M3\_P5 DOCUMENTATION OF BRAIN MEASURES 20260910

Information regarding MRI neuroimaging procedures and the processing of T1-weighted and diffusion-weighted scans. As well as information about the atlas-based extracted measures and brain-based aging algorithms

### M3\_P5 DOCUMENTATION OF CANTAB 20260910

Information about the CANTAB cognitive assessments

### M3\_P5 DOCUMENTATION OF PSYCHOPHYSIOLOGY 20260910

Information regarding psychophysiological methods for data collection during the psychophysiology and neuroimaging visits, and subsequent data processing

### M3\_P5 DOCUMENTATION OF SCALES 20260910

Documentation of self-report/questionnaire measures

### M3\_P5 RESTRICTED ACCESS 20260106

Overview of additional data available through restricted access procedures

*Note: this document is NEW for the M3 P5 February 2026 data sharing update.*

### M3\_P5 ACKNOWLEDGEMENT TEXT 20230403

A sample acknowledgment text to be included in publications utilizing this data is available

## E. Where Can I Access the Restricted Raw Imaging, Psychophysiological, and Behavioral Data?

Access to the MIDUS 3 raw MRI data (structural, task functional, resting state functional, diffusion-weighted imaging, and resting arterial spin labeling perfusion) as well as the raw and trial-by-trial psychophysiological and behavioral data.

For an overview of what data is available see M3\_P5\_RESTRICTED\_ACCESS\_20260106 and <https://midus.wisc.edu/midus-neuroscience-repository/> for instructions on how to access.

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Please report any errors or inconsistencies you find in the data or documentation to  
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